

23CSE111
OBJECT ORIENTED PROGRAMMING
LAB SESSION

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Question-1

Java Program to swap two numbers

CODE:

```
public class swap
{
    static void swapval(int m,int n)
    {
        int temp=m;
        m=n;
        n=temp;
        System.out.println("Value of m is "+m+" and value of n is "+n);
    }
    public static void main(String[] args)
    {
        int m=9,n=5;
        swapval(m,n);
    }
}
```

OUTPUT:

```
C:\Users\ThinkBook\Downloads>javac swap.java

C:\Users\ThinkBook\Downloads>java swap
Value of m is 5 and value of n is 9
```

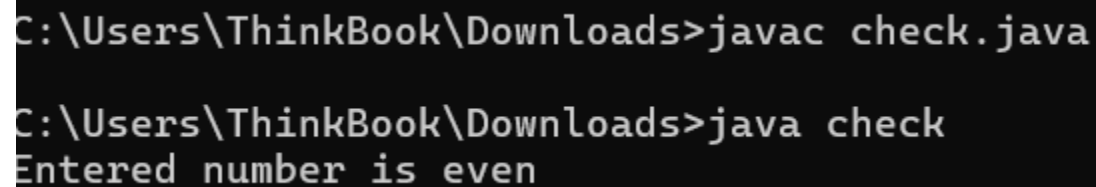
QUESTION-2

JAVA Program to check whether a given number is even or odd.

CODE:

```
public class check
{
    public static void main(String[] args)
    {
        int num=10;
        if (num%2==0)
        {
            System.out.println("Entered number is even");
        }
        else
        {
            System.out.println("Entered number is odd");
        }
    }
}
```

OUTPUT:



```
C:\Users\ThinkBook\Downloads>javac check.java
C:\Users\ThinkBook\Downloads>java check
Entered number is even
```

QUESTION-3

JAVA Program to find the perimeter of a rectangle

CODE:

```
public class perimeterrect
{
    static void perimeter(int length,int breadth)
    {
        int perimeter=2*(length+breadth);
        System.out.println("The perimeter of the given rectangle of length
"+length+" and breadth "+breadth+" = "+perimeter);
    }
    public static void main(String[] args)
    {
        int length=10;
        int breadth=20;
        perimeter(length,breadth);
    }
}
```

OUTPUT:

```
C:\Users\ThinkBook\Downloads>javac perimeterrect.java
C:\Users\ThinkBook\Downloads>java perimeterrect
The perimeter of the given rectangle of length 10 and breadth 20 = 60
```

QUESTION-4

JAVA Program to find armstrong numbers between 2 numbers

CODE:

```
public class armstrong {
    static void ArmstrongNum(int l, int h)
    {
        for (int j = l + 1; j < h; ++j) {
            int y = j;
            int N = 0;
            while (y != 0) {
                y /= 10;
                ++N;
            }
            int sum_power = 0;
            y = j;
            while (y != 0) {
                int d = y % 10;
                sum_power += Math.pow(d, N);
                y /= 10;
            }
            if (sum_power == j)
                System.out.print(j + " ");
        }
    }
    public static void main(String args[])
    {
        int n1 = 50;
        int n2 = 500;
        ArmstrongNum(n1, n2);
        System.out.println();
    }
}
```

OUTPUT:

```
C:\Users\ThinkBook\Downloads>javac armstrong.java
```

```
C:\Users\ThinkBook\Downloads>java armstrong  
153 370 371 407
```

QUESTION-5

JAVA Program to determine LCM of 2 numbers

CODE:

```
public class LCM {  
    public static void main(String[] args)  
    {  
        int a = 15, b = 25;  
        int ans = (a > b) ? a : b;  
        while (true) {  
            if (ans % a == 0 && ans % b == 0)  
                break;  
            ans++;  
        }  
        System.out.println("LCM of " + a + " and " + b  
            + " : " + ans);  
    }  
}
```

OUTPUT:

```
C:\Users\ThinkBook\Downloads>javac LCM.java  
  
C:\Users\ThinkBook\Downloads>java LCM  
LCM of 15 and 25 : 75
```

QUESTION-6

JAVA Program to find the largest numbers

CODE:

```
public class largest {  
    static int biggestOfThree(int x, int y, int z)  
    {  
  
        return z > (x > y ? x : y) ? z : ((x > y) ? x : y);  
    }  
    public static void main(String[] args)  
    {  
        int a, b, c;  
        int largest;  
        a = 5;  
        b = 10;  
        c = 3;  
        largest = biggestOfThree(a, b, c);  
        System.out.println(largest+ " is the largest number.");  
    }  
}
```

OUTPUT:

```
C:\Users\ThinkBook\Downloads>javac largest.java  
  
C:\Users\ThinkBook\Downloads>java largest  
10 is the largest number.
```